Cross-County, Chapel Hill, North Carolina

On the Orange–Durham county line, the winter danger is not the storm itself but day three of a power outage, when a family brings a grill or generator inside to keep warm. The single most important step is a working carbon monoxide alarm, installed before December, in every rented and non-English-speaking household. That protects people in any winter, El Niño or not.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Orange County and Durham County, NC (inferred; the neighbourhood straddles the line)	None published for the US, North Carolina, or either county	n/a here; global 6-month figure 239,000 ± 11,000	Ice-storm outages lasting days, the carbon monoxide poisoning that follows, and cold exposure. Local flash flooding is secondary.

10/10 recommendations checked. 6 corrected, 1 unverified (renters' flood cover), 2 contraindicated (“El Niño ice season” messaging; $800 battery backups as oxygen protection), and 1 out of scope here (Camelot Village-style buyouts). Exposure figure: Climate Impact Lab, September 2026, pre-peer-review, heat only. No figure exists at this resolution, and this winter's hazard here is cold, not heat, so no number is borrowed. **The**

EXECUTIVE SUMMARY — AT A GLANCE



35.96°N 79.04°W, United States of America · Natural Earth

- **The risk.** The winter danger is day three of a power outage, when a family brings a grill or generator inside: the December 2002 ice storm left 24 dead statewide and some without power for up to 10 days.
- **First action.** Durham Fire, Chapel Hill Fire, and Orange County Emergency Services install sealed 10-year CO alarms in rented and non-English-speaking households by 30 November.
- **The money.** $70k floor, $110k target, $205k ceiling; the floor pays $25k for CO alarms and $45k for utility help.
- **Do not.** Do not buy $800 battery power stations as the way to protect home-oxygen users: a unit that size runs a 300 W concentrator for about three hours.*

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* Suggested formulation: check with a health worker before sharing.

funding target fell 68% in verification, from $345k to $110k. A cited benefit-cost ratio did not exist in its source, one placement was contraindicated, and two duplicated money that is already committed.

Where this is. “Cross-County” is a real-estate neighbourhood label (Redfin; about 764 residents), not a municipal one. Census geocoding puts 1547 Fountain Ridge Rd in Orange County and the Town of Chapel Hill (tract 121.01), and Portofino Pl in Durham County and the City of Durham (tract 20.33). Pope Rd and Clark Lake Rd are on the Durham side. Mailing addresses mix Chapel Hill 27517 and Durham 27707. **Every action below therefore has an Orange half and a Durham half**, with different fire departments, social services, shelters, and emergency alert systems. The area drains to tributaries of New Hope Creek, not to the Bolin and Booker creek corridor where Chapel Hill's well-known floods happen.

WHAT THIS PLACE SHOULD DO — RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. One rule underlies every action: in central North Carolina, ice storms kill on days three to six of an outage, through carbon monoxide and cold. Prepare for that in any winter. A forecast is not the reason to act.

1. Install sealed 10-year carbon monoxide alarms in rented and non-English-speaking households before 30 November, and say “no grill, no generator indoors” in Spanish, Karen, and Burmese

PARTIALLY CONFIRMED | ALL SEVEN LENSES | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Durham side: Durham Fire Department (free “Alarm for Life” installs). Orange side: Chapel Hill Fire Department with Orange County Emergency Services. Outreach: Refugee Community Partnership, El Centro Hispano	Installs by 30 Nov; voice notes circulating by 15 Nov	Durham side free; ~$25k for ~500 Orange homes at ~$50 installed (retail-based estimate)	Carbon monoxide deaths in the first days of an outage

* Suggested formulation: check with a health worker before sharing.

This is the clearest local evidence in the record. After the December 2002 ice storm, Mecklenburg County recorded 124 symptomatic CO poisonings over 4–13 December. **25 of the 26 severe cases (96.2%) were in homes with no working CO alarm**, and 56.3% of cases over age five did not speak English (CDC *MMWR* 53(09), 2004: a surveillance report, not peer-reviewed). Duke University Hospital's emergency department saw more than 200 CO patients in the same storm, 18 needing hyperbaric oxygen, and most of them were Latino (Ghim & Severance, *South Med J* 2004). In Wisconsin (2014–16), poisoning cases with no detector had 3.2 times the odds of those where a detector alarmed (95% CI 1.5–6.9; observational, poison-centre records). Outreach should travel as two-minute voice notes passed along on WhatsApp and Viber by the people newcomers already rely on, because TV and phone warnings stop when the power does.

Corrections: the Mecklenburg data shows that *renting and language* were the risk markers. All-electric homes without garages had a slightly lower rate (RR 0.7, not significant), so target renters and language, not heating type. A legal gap does remain: NC G.S. 42-42(a)(7) requires landlords to fit CO alarms only where there is a fossil-fuel appliance, a fireplace, or an attached garage. The “7.2:1 benefit-cost ratio” one lens cited is not in its source (Hampson 2015 gives a break-even instead), and it has been dropped. The “Hispanic residents were 65% of CO poisonings” figure could not be found in a primary source. Durham Fire already installs alarms free, so on that side the action is referral, not a new program. Habitat for Humanity of Orange County does about 25 repairs a year and cannot install at this scale. Transferability is strong: same state, same hazard, same storm.

2. Build an outage welfare-check roster of people who are 65 or older, live alone, or depend on powered medical equipment, pair each with a neighbour, and staff checks for days 1–6

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH · FRONTLINE · SKEPTIC · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Orange County Department on Aging (Senior Resource Teams, Volunteer Connect 55+); Orange County Emergency Services and Durham County Emergency Management; county health departments (emPOWER)	Roster and pairings by 1 Dec	Staff and volunteer time; $15k for a call-tree coordinator, conditional on a named operator	Hypothermia, CO, and device-failure deaths among people nobody checks on after the first night

* Suggested formulation: check with a health worker before sharing.

Build the roster from the HHS emPOWER map (ZIP-level counts are public; named lists need a health-department request), county outreach lists, and a door-to-door canvass. Each check covers three things: heat, a CO source in the house, and oxygen or device power. When ice is forecast, ask dialysis units to bring treatment forward: early dialysis before Hurricane Sandy was associated with lower 30-day mortality (OR 0.72, 95% CI 0.52–0.997; Lurie et al., *AJKD* 2015). The 2003 New York blackout raised non-accidental deaths by 25% (95% CI 12–41%; Anderson & Bell, *Epidemiology* 2012).*

Corrections and transferability: both studies come from summer or hurricane events, not ice storms. Winter deaths follow a different pattern. In Texas in February 2021, 158 of 246 deaths were hypothermia and at least 19 were CO (Texas DSHS), so checks must cover warmth and generators, not just devices. Duke Energy's "Medically Essential" program notifies enrolled customers, but its own page says enrolment "does not mean that you will be the first to have power restored." Do not tell people it does. The Senior Resource Team programs are real. No existing outage check-in program was found, so the $15k is unverified until an operator commits. The emPOWER counts for ZIPs 27517 and 27707 were not retrieved.

3. Make sure every home-oxygen user has backup cylinders from their supplier and a refill plan for a multi-day outage

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · PHILANTHROPIST (AS CORRECTED) | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County health departments with durable-medical-equipment suppliers; roster volunteers check during action 2	**Before 1 Dec**	**Normally covered by the supplier; ceiling $10k only if a delivery gap is shown**	**Oxygen failure when a concentrator loses power for days**

This replaces the original proposal to buy $800 battery backups (see *Do not do this*). A concentrator draws about 300 W. Supplier cylinders cover days, while a consumer battery covers hours, and backup cylinders are the standard of care in ASPR TRACIE's utility-failure guidance for oxygen users.*

* Suggested formulation: check with a health worker before sharing.

Cylinder logistics in this area (who delivers, and how fast after an ice storm) were not verified from a primary source.

4. Before 1 November, enrol eligible households in winter shutoff protection, open crisis heating assistance now, and have LIEAP documents ready

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Orange County DSS and Durham County DSS (certification); households call Duke Energy; navigators from Department on Aging, IFC, El Centro	Protection runs 1 Nov – 31 Mar; enrol from now	Staff time; plus $45k top-up for utility help in Orange, where the Share the Light line is $0	Households going into an ice storm already without heat

Between 1 November and 31 March, NC utilities may not disconnect a household that meets three tests: someone in it is 65 or older or disabled, it cannot pay under an installment plan, and the county DSS certifies it as eligible for energy assistance. **It is not automatic**; the household must call the utility and send the forms (NC 211; NCDHHS DSS-8117). Federally funded SFY2026–27 estimates: Orange County has Crisis Intervention Program (CIP) $450,968 and LIEAP $567,372; Durham County has CIP $1,165,467 and LIEAP $1,466,298. CIP runs year-round, so applications can start today. Last winter LIEAP opened late, on 10 December for people 60 and over or disabled and on 2 January for everyone else, because federal money arrived late. Plan for that to happen again.

Corrections: LIEAP cannot be pre-filed; the action is to have documents ready. Duke's Share the Light Fund ($6.4M across all Duke territories, year to date at 17 Dec 2025) is budgeted at only $11,777 for Durham and $0 for Orange in SFY27. Orange County's Emergency Housing Assistance ($4,500 cap, about 12 households a month) is rent-focused and has no top-up mechanism, so the utility money

* Suggested formulation: check with a health worker before sharing.

should go through Orange DSS Emergency Assistance (currently $200 per 12 months) or IFC instead. **The verifiers disagreed on which Duke subsidiary serves this area**: Duke Energy Carolinas (Chapel Hill's system was sold to Duke Power in 1977) or Duke Energy Progress (named in the DSS budget lines). Check the bill. The rule number (NCUC R12-11) could not be fetched.

5. Close the gaps in the existing White Flag shelters: transport for people without cars, announced 48 hours ahead, and a hotel overflow for outage nights

PARTIALLY CONFIRMED | EMERGENCY MGR · SKEPTIC · FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Orange side: Orange County with IFC (Community House, 1315 MLK Blvd; HomeStart, 2505 Homestead Rd; White Flag site, 828 MLK). Durham side: City of Durham with Urban Ministries of Durham and Entry Point	Transport and overflow contracts signed by 1 Dec; released on an NWS Ice Storm Warning	~$25k contingency: 20 rooms × 10 nights × $125	Cold exposure among people who cannot reach a shelter, and homeowners with no heat on day 4

The shelters already exist. Orange County opens White Flag shelters at a forecast of 32°F or below, and IFC runs Cold Weather Cots every night from November to March. Durham opens its White Flag shelters at 32°F, or 35°F with precipitation. Street Outreach takes calls until 6pm (919-886-3351), and buses are fare-free. What is missing is cover for *housed* people who lose power, and for people who cannot get to a site: an older homeowner in a cold ranch house on day four. In Austin in February 2021, warming-centre shuttles failed because people heard about the pickup points too late (Austin Chronicle; no after-action report was found).

* Suggested formulation: check with a health worker before sharing.

Corrections: no study measuring how much warming centres reduce cold deaths was found. This action rests on plausibility and on the Texas death pattern, not on an effect size. Orange County's unsheltered count rose 45% in 2025 (point-in-time count; small numbers). The lens's original $100k for 50 nights was cut, because cots already run all season and $100 a night is a hotel rate.

6. Mark and barricade local low crossings on the New Hope Creek tributaries, and tie door knocks to NWS flash-flood warnings

PARTIALLY CONFIRMED · DEMOTED | EMERGENCY MGR · HISTORIAN · SKEPTIC | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Durham Fire Department and Chapel Hill Fire Department; NCDOT and city/town public works for barricades	**Crossings identified by 15 Nov**	**Staff time; barricade costs not quantified**	**Drowning in vehicles at flooded crossings**

Over half of US flood deaths are in vehicles (NWS, about 90 deaths a year). Tropical Storm Chantal (6–7 July 2025) killed 6 people in North Carolina, with confirmed vehicle deaths in Chatham, Orange, and Alamance counties. Chapel Hill crews made more than 50 rescues. Barricades and marking do more than slogans: an LSU study of Texas found no significant drop in vehicle flood deaths after “Turn Around, Don't Drown” began.

Corrections: **demoted, because Cross-County is not on the documented flood path.** No Chantal report names its streets. Durham's 80+ rescues were in Old Farm, in north Durham on the Eno River, about ten miles away. The USGS gauge one lens proposed (02097386, Booker Creek) is in the wrong watershed. A secondary dam database lists Clark Lake Dam (NC01048), a private dam on a New Hope Creek

* Suggested formulation: check with a health worker before sharing.

tributary, as “significant hazard” with no Emergency Action Plan and a last inspection in 2002. This could not be confirmed in the federal National Inventory of Dams. NCDEQ Dam Safety should be asked to check it, and if it holds, the owner should be asked to write an Emergency Action Plan. Colony Woods Dr has recorded stormwater flooding (Chapel Hill staff report, 2002).

7. Check each low-lying address on the state flood maps before selling renters' contents-only flood cover, and buy by about 8 November if it is needed

UNVERIFIED | ECONOMIST · HISTORIAN | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
Households and landlords; Town of Chapel Hill and City of Durham floodplain staff	Policy bought by ~8 Nov is in force by ~8 Dec	Premium not quoted	Uninsured contents loss for renters (financial harm, not deaths)

NFIP's 30-day waiting period is set by statute. At Camelot Village in 2013, renters found that their policies excluded flood.

It is unverified whether any Cross-County address is in a Special Flood Hazard Area. Check each address on NC FRIS (fris.nc.gov) before promoting this action. If no address is in or next to a flood zone, drop it.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Durham Fire	Chapel Hill Fire	Orange Co. Emergency Services	Durham Co. EM	County health depts	Orange Dept on Aging	Orange / Durham DSS	IFC / Urban Ministries	Duke Energy	RCP / El Centro	Residents
1. CO alarms; multilingual message	**A/R (Durham side)**	**A/R (Orange side)**	**R**	**C**	**C**					**R**	**R**
2. Welfare-check roster, days 1–6			**C**	**C**	**R**	**A/R**			**I**	**C**	**R**
3. Backup oxygen cylinders					**A/R**	**R**					**R**
4. Winter shutoff protection, CIP, LIEAP						**R**	**A/R**	**R**	**R**	**R**	**R**
5. White Flag gaps: transport, overflow			**A (Orange side)**	**A (Durham side)**		**C**		**R**		**I**	
6. Low crossings; dam EAP check	**A/R**	**R**	**C**	**C**							**R**
7. Flood-map check; renters' cover			**C**	**C**							**A/R**

What the chart shows. **No action has a single owner across the whole neighbourhood.** Five of the seven split at the county line, so rows 1 and 5 carry two accountable parties, one per side. **Accountability for action 2, the roster, sits with the Orange County Department on Aging, which has no Durham counterpart named here.** Before December, someone has to decide who checks on the Durham-side households.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs. No placement carries a measured cost per death averted; amounts are bottom-up unit costs.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$70k	**$110k**	**$205k**	**Whether an ice storm comes at all; El Niño makes this winter wetter, not reliably icier**

Floor: CO alarms $25k + utility help $45k = $70k. **Target:** $25k + outage overflow $25k + utility help $45k + call tree $15k + batteries $0 = $110k. **Ceiling:** CO alarms $40k + overflow $50k + cylinder-gap contingency $10k + utility help $90k + call tree $15k = $205k. **The target fell 68% in verification, from $345k.** The verifiers dropped the $80k battery placement (contraindicated), cut the shelter overflow from $100k to

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$25k (cots already run all season), cut CO alarms from $60k to $25k (Durham Fire installs them free; Habitat cannot install at scale), and cut and moved the utility help from $90k to $45k (it does not fit the rent-focused Emergency Housing Assistance program). These are town and county-scale amounts: Cross-County itself is only a few hundred households. Spending for the Durham half should go through Durham DSS and Urban Ministries of Durham, not through Orange County agencies.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Chapel Hill Fire Department / Orange County Emergency Services	Action 1: sealed CO alarms in ~500 Orange-side homes	$25k (was $60k)	Pre-positioned stock + direct grant, by 1 Nov	500 × ~$50 (retail-based; no bulk quote found). MMWR 2002: 25 of 26 severe cases in homes with no alarm. No published cost per death averted	PARTIALLY CONFIRMED
Orange DSS Emergency Assistance or IFC emergency assistance	Action 4: utility arrears and heating help	$45k (was $90k)	Top-up	75 households × $600; Orange's SFY27 Share the Light line is $0. Needs agency agreement	PARTIALLY CONFIRMED
IFC (2024 expenses $3.7M) with Orange County	Action 5: hotel overflow on outage nights	$25k (was $100k)	Contingency grant released on an NWS Ice Storm Warning	20 rooms × 10 nights × $125. No study measures warming-centre effect	PARTIALLY CONFIRMED
Orange Department on Aging / CERT	Action 2: outage call tree	$15k	Direct grant, if an operator commits	One season's coordination; no existing check-in program found	UNVERIFIED
Department on Aging + Piedmont Health	~100 battery backups at $800	$0 (was $80k)	Dropped	~1 kWh runs a 300 W concentrator for ~3 hours; suppliers provide cylinders	CONTRAINDICATED

The funder's one thing: the cheapest death to prevent here is a CO poisoning in the first days of an outage, and no budget line pays for the alarm before it. Release the alarm money now, unconditionally. If no ice storm comes, the alarms stay on the walls for ten years. Hold the shelter money on a written trigger, an NWS Ice Storm Warning. The single most valuable step costs nothing: a DSS caseworker certifying an older household for winter shutoff protection before 1 November.

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Advice that circulates for this hazard and would cost lives, money, or trust here.

Build the campaign around “El Niño means an ice season”

CONTRAINDICATED

The link is weak. The NC State Climate Office ties El Niño to a *wetter* winter with an active southern storm track. Whether that falls as ice depends on cold-air outbreaks, which El Niño can suppress. In the strong 2015–16 El Niño, the office forecast above-normal snow. The winter was instead the 4th-wettest and 16th-warmest on record, with below-normal snowfall at Raleigh, Greensboro, and Charlotte, and the office says it “underestimated” the southern jet. Ice storms come in any phase: the December 2002 storm fell in a moderate El Niño (ONI +1.2), but Winter Storm Fern (January 2026, 0.25–0.5 in of ice forecast) fell in a weak La Niña. A campaign sold on El Niño will look wrong in a mild winter, and that costs trust just before the next La Niña ice storm.

Instead: “Ice storms hit here in any winter. Get ready once, for every winter.” Use this El Niño as the occasion to act, not the reason.

Buy $800 battery power stations as the way to protect home-oxygen users *

CONTRAINDICATED

An $800 unit stores roughly 1 kWh. That runs a 300 W concentrator for about three hours, less in the cold, and the battery cannot be recharged during a multi-day outage. The 2002 ice storm took up to ten days to restore. A battery that gives a family false confidence on day one leaves the oxygen user without oxygen on day two.

Instead: backup cylinders from the oxygen supplier and a refill plan (action 3). Use batteries only as a short bridge, and then they need 1.5–2 kWh pure-sine-wave units at about $1,000–1,900.

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Out of scope here, not contraindicated: owner-by-owner FEMA buyouts on the Camelot Village model. They are a sound lesson for the Bolin Creek corridor, several miles north-west, where federal buyout money was returned in 2005–09 because every owner had to agree. No comparable flood-prone complex is known in Cross-County.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1982–83, 1997–98	Strong El Niño winters	Nothing specific to this place was found	—
Jan 2000	Record snowstorm, 20.3 in at RDU, badly forecast	Roads unplowed for days; hospitals needed National Guard generators	Not established
Dec 2002	Ice storm, up to 1 in of freezing rain; moderate El Niño (ONI +1.2)	24 dead statewide; up to 10 days without power; 200+ CO patients at Duke's emergency department, 18 hyperbaric; a 31-year-old Durham man died of CO	State law now requires CO alarms in rentals, but only where there are fossil-fuel appliances, fireplaces, or attached garages
Jun 2013	Bolin Creek flash flood, Camelot Village (not Cross-County's watershed)	68 units flooded, ~$1.5M damage; Town paid ~$412k of its own repairs	Buyout attempt failed again for lack of all-owner consent
2015–16	Strong El Niño	4th-wettest and 16th-warmest winter; snowfall below normal (State Climate Office)	The office acknowledged that it underestimated the southern jet
Jul 2025	Tropical Storm Chantal (not on Cross-County's drainage)	6 dead in NC; 50+ Chapel Hill rescues; Eastgate damaged; 80+ Durham rescues in Old Farm	Booker Creek storage basins remain on hold
Jan 2026	Winter Storm Fern; weak La Niña	0.25–0.5 in of ice forecast for Orange County; state of emergency 22 Jan	—

The recurrence nobody has noticed: the deaths that follow a Triangle ice storm are mostly carbon monoxide, not cold, and they fall on renting and non-English-speaking families who improvise heat on day three. That was documented in 2002. Twenty-four years on, the alarm law still

* Suggested formulation: check with a health worker before sharing.

covers only rentals with fossil-fuel appliances, fireplaces, or garages.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **“El Niño makes this a wintry season”** (State Climate Office's 2015 forecast, the Historian) versus **“the ice signal is weak and the record shows ice in any phase”** (the Skeptic). Verification sided with the Skeptic on messaging. What would settle it: a frozen-precipitation-day count by ENSO phase for RDU, which nobody has published in peer review.
- **“All-electric homes are the gap, because the law exempts them”** (Public health) versus **“renters and language predicted poisoning, not heating type”** (the 2002 MMWR data, via verification). Both point to renters. The disagreement is about whether to also lobby to close the statutory gap, which cannot happen before this winter.
- **Warming centres as the largest funding placement** (Philanthropist, $100k) versus **“no evidence they reduce deaths, and people don't reach them”** (Public health, Skeptic). Resolved for now as a $25k contingency aimed at transport and overflow. An evaluation of Orange County's White Flag use would settle it.
- **Which Duke Energy subsidiary serves this area.** Two verifiers disagreed: Carolinas or Progress. It changes which disconnection data applies and how Share the Light is routed. Settle it from a bill.

WHAT NOBODY COVERED — BLIND SPOT

Falling trees, roads, and the county line itself. No lens examined deaths from falling limbs and vehicle crashes on ice, which are common in Southern ice storms, or the wooded lots typical of this area. No lens checked which emergency alert system a given Cross-County street receives. Orange County and Durham County run separate systems, and a household on the line can register with the wrong one or neither. A ten-minute check, sign up for both, may be the cheapest action in this briefing, and it has not been verified.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

* Suggested formulation: check with a health worker before sharing.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people. They are not neighbourhood figures. The El Niño projection covers *heat only*; ice, cold, carbon monoxide, and flood are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The location was taken as given: Cross-County, Chapel Hill, North Carolina. Climate Impact Lab impact region, as the report records it: Orange County and Durham County, NC (inferred, not looked up in the Impact Map; the neighbourhood straddles the line, confirmed by Census geocoding of two addresses). Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not the winter hazard here, so the historical record carries the hazard instead of a borrowed number.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief. Four of the seven independently flagged that the neighbourhood is on the Durham side of the label.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted in an action's favour. All seven lenses cited the 2002 ice-storm CO cluster, but that evidence is local (same state, same hazard), so the convergence was allowed to stand. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in this jurisdiction, a named payer, a trigger threshold, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.

* Suggested formulation: check with a health worker before sharing.

- **Adaptation Inventory.** Not consulted. A verifier fetched the page in this run and it rendered empty (0 entries) without its interactive interface, and the download sits behind a form. Earlier runs on 8 October 2026 that drove the US filter in a browser found 19 US entries, none covering CO alarms, outage welfare checks, warming centres, or energy assistance. Every action is tagged “not consulted”, which is not the same as “no precedent found”.
- **Health wording.** Passages where a wording error could hurt someone (oxygen, dialysis, and medical equipment in an outage, and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so. The Town's ~$412k self-paid 2013 repair cost is a shown subtraction ($679,576 – $267,092). Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. “Cross-County” is a Redfin label with no official boundary. Only two addresses were geocoded. The Duke Energy subsidiary is unresolved. Duke's Medically Essential page blocked direct fetch. The NCUC rule text returned 403. emPOWER counts for 27517 and 27707 were not retrieved. Clark Lake Dam rests on a secondary database and was not found in a search of the federal National Inventory of Dams. FEMA flood-zone status was not checked for any address.

References · verified by Phase 5 unless marked. CDC MMWR 53(09), CO poisoning after the 2002 NC ice storm (corrected) · Ghim & Severance 2004, Duke ED CO cases (corrected) · CO detector effectiveness, Wisconsin 2014–16 (verified) · Hampson 2015, CO alarm cost analysis (corrected: no 7.2:1 ratio) · NC G.S. 42-42 (verified) · Durham Fire community programs (verified) · Durham Fire home-safety request (Spanish) (verified) · 2002 NC ice storm overview (lens) · Durham CO death, 2002 (lens) · NOAA CPC ONI v6 (verified) · NC State Climate Office, 2015–16 winter recap (verified) · Chapelboro, Winter Storm Fern (verified) · AMS 2000 abstract, ENSO and NC winter storms (lens) · Lurie et al. 2015, early dialysis (verified) · Anderson & Bell 2012, NYC blackout mortality (verified) · Texas DSHS, February 2021 storm deaths (verified) · HHS emPOWER (verified; counts not retrieved) · Duke Energy Medically Essential (via search summary) · Orange County Senior Resource Teams (verified) · Oxygen concentrator battery runtime (verified) · ASPR TRACIE, oxygen utility-failure tip sheet (verified) · NC 211, winter termination protection (verified) · NCDHHS Orange County SFY2027 estimates (verified) · NCDHHS Durham County SFY2027 estimates (verified) · NCDHHS, LIEAP 2025–26 dates (verified) · Duke Energy, Share the Light 2025 (corrected) · WUNC, Duke disconnections (verified; Carolinas data) · Orange County Emergency Housing Assistance (verified) · Orange County DSS Emergency Assistance (verified) · Governor's pre-storm state of emergency, 9 Jan 2025 (verified) · Orange County White Flag shelters (verified) · Orange County Cold Weather Cots (verified) · Durham winter shelters (verified) · Axios Raleigh, Orange unsheltered count (corrected) · Austin Chronicle, warming-centre shuttles (partly verified) · IFC, IRS filings (verified) · Redfin, Cross-County neighbourhood (lens) · US Census geocoder (queried 9 Oct 2026) · DCHC MPO, Pope Rd (verified) · Clark Lake Dam (secondary database) (unverified) · Chapel Hill staff report, Colony Woods Dr, 2002 (verified) · NHC Tropical Cyclone Report, Chantal (verified) · Chantal rescues and Eastgate damage (verified)

* Suggested formulation: check with a health worker before sharing.

· Indy Week, Old Farm after Chantal (verified) · USGS 02097386, Booker Creek (verified; wrong watershed) · NWS, Turn Around Don't Drown (verified) · LSU thesis, Texas vehicle flood deaths (verified) · Indy Week, Camelot Village 2013 (verified) · Daily Tar Heel, Town Hall flood costs (lens) · Chapelboro, Camelot buyout (verified) · ABC11, January 2000 snowstorm (lens) · Town of Chapel Hill, Elliott Road flood storage (lens) · Chapelboro, Karen community (lens)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 9 October 2026 for Cross-County, Chapel Hill, North Carolina. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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